THE UNITED REPUBLIC OF TANZANIA
COMPETENCE BASED CURRICULUM
JOINT EXAMINATIONS PRIVATE AND GOVERMENTS SCHOOLS-TANZANIA (JEPGOS-T)
FORM FOUR MOCK EXAMINATION -2025

BIOLOGY 01

(For Both School and Private Candidates)

Time 3h : 00min **August 2025**

INSTRUCTION.

1. This paper consist of section A, B and C with total of eleven (11) questions.
2. Answer all question in section A and B and two question from section C
3. Section A carries 16 marks section B carries 54 marks and section C carries 30 marks
4. All writing should be in blue or black ink except for diagrams that must be draw in pencil
5. Remember to write your candidate examination number at the top of every page.
6. Cellular phones and any unauthorized materials are not allowed in the examination room.

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SECTION A (16 MARKS)

ANSWER ALL QUESTIONS

1. For each of the following items (i- x) choose the correct answer among the given alternative and write its letter besides the item number on the answer sheet provided.

   i. A veterinarian treating a dog for a skin infection consider potential causes. Which rank of classification would be most responsible or often most specific information about potential parasites that could be responsible for the infection?

      A. Order
      C. Genus
      B. Family
      D. Class

   ii. A school composting bin filled with discarded fruits and vegetable mainly contains

      A. Non- living organic matter
      B. Living micro- organism
      C. A mixture of living and non- living components
      D. Only inorganic chemicals

   iii. Which protein is found on the surface of pathogens that cause infections and diseases

      A. Antibodies
      C. Rhesus factors
      B. Antigens
      D. Vectors

   iv. Mass flow occurs in one of the following system

      A. Lymphatic system
      C. Respiratory system
      B. Reproductive system
      D. Blood group system

   v. A land scape architect designing a park chases to plant ever green trees along a noisy highway. This decision aims to utilize the plant ability to

      A. Produce edible fruits for park visitors
      B. Provide shade and shelter for wild life
      C. Purity the air by removing moving harmful pollutants
      D. Reduce noise level through sound absorption by leaves

   vi. A man with Brow iris eyes married a woman with blue iris eyes in F1 generation all of their offspring were brown iris eyes. What does this suggest about brown iris colour?

      A. Incomplete dominance
      B. Complete dominance

C. Co- dominance
D. Multiple allelism

vii. Which of the following types of protein filaments made up muscles?

A. Adenine and guanine
C. Adenine and thymine
B. Ligaments and tendons
D. Actin and myosin

viii. All substance carried by the blood towards the kidney are filtrated during urine formation but the following food substance is not one of the glomerulus filtrate

A. Glucose
C. Water
B. Protein
D. Amino acids

ix. The growth of human being starts before and after birth by following a systematic growth pattern which is

A. Localized growth
C. Allometric growth
B. Isometric growth
D. Diffused growth

x. Veins are blood vessels that blood circulation is done through it. As a biologist what feature among the following you have to use to identify veins.

A. Presence of valve
C. Absence of valve
B. Presence of blood with high pressure
D. Large muscular wall.

2. Match the items in list A corresponding to list B by writing the letter which is corresponding to the item.

LIST A	LIST B
i. The change in allele frequencies that occurs overtime within a population	A. Natural selection
ii. The change in the relative frequencies in which a gene variant (allele) occurs in a population due to random sampling	B. Biogenesis
iii. The process where living organism arise from pre-existing organism	C. Selection pressure
iv. Agent of differential mortality or fertility had tend to make a population change genetically	D. Cosmology
	E. Divergent evolution
	F. Convergent evolution
	G. Abiogenesis
	H. Microevolution
	I. Adaptive radiation

v. The evolution of an animal or plants group into a wide variety of type adapted to specialized mode of life. vi. The science of the origin and development of the universe.	J. Genetic drift

SECTION B (54 MARKS)

3. (a) Ringing in plants involve the removal of the bark of trees. This hinder the communication between the upper and lower part of the plants. In four points briefly elaborate the impacts of ringing on the stem of the hibiscus flower. **(04 marks)**

   (b) Xylem and phloem are vascular bundles in plants differentiate them by giving five points **(05 marks)**

4. (a) Asha was suffering from Diabetes mellitus when she went for medical treatment was advised to reduce intake of carbohydrate (glucose) in order to reduce the amount of glucose in the blood stream. What are the hormones involved in the blood sugar balance? **(01 mark)**

   (b) For each hormone you have mentioned in 4 (a) above state cells where the hormones named are produced**. (01 marks)**

   (c) With help of equation explain the mechanism that take place when the hormones named above are involved in regulation of glucose. **(01 marks)**

5. (a) (i) Use an example to explain why the common names of organism are sometimes confusing or misleading **(1 ½ marks)**

   (ii) How can using scientific names help to avoid the confusion? **(1 ½ marks)**

   (b) Outline the rules to be followed when naming an organism (give four points) **(04 marks)**

   (c) Viruses are the boundary between living and non- living things. Discuss by giving two points in each case. **(02 marks)**

6. (a) Explain the following phenomena as used in biology.

(i) Insulin and glucagon hormone work antagonistically **(01 marks)**

(ii) In cold and humid weather human usually produce large quantity of urine that is colourless **(01 mark)**

(iii) Fish dies when taken outside water bodies **(01 mark)**

(iv)When plant seedling placed horizontally on ground for some days, the root grow down while the shoot grow upward. **(01 mark)**

(v) People with diabetes usually experience thirsty and frequent urination (**01 mark)**

(b) Juma was narrating a story in a class about a person who has unbelievable strength and energy when subjected in an extremely dangerous situation. What is the chemical released in Juma's body during that situation and give its three roles. **(04 marks)**

7. (a) Explain the problems faced by albino under the following guidelines.

   (i) Vision **(01 mark)**

   (ii) Exposure to light **(01 mark)**

   (b) In a legal dispute two families of John and Joseph both claimed to have sired a girl by the name Leila whose blood group was determined to be AB. John was established to have blood group A and his wife blood group O Joseph had blood group A and his wife has Blood group B . Determine which family is likely biological parents of Leila. Show your working **(7 marks)**

8. (a) Your Community is implementing new recycling program how can you effectively sort different types of waste for proper recycling? Should you focus on material type (plastic paper benefit e.t.c) colour, size? What are the benefits and drawbacks of each method? **(03 marks)**

   (b) You start a compost bin to recycle kitchen scraps after a weeks you see the pile shrinking and turning into dark, crumby compost, explain how experiments on aerobic respiration can help to explain this transformation. **(06 marks)**

SECTION C (30 MARKS)

Answer only two questions.

9. At Mtakuja Village people have a belief that only those who are seriously sick should visit hospital for VCT. This belief is totally wrong, use six points to show the wrong side of this belief. **(15 marks)**

10. One of the investigation of disadvantages of protozoa students discovered a single celled parasitic protozoa found in phylum apicomplexa which cause an endemic diseases to man. This disease can lead to inflammation and rupture of spleen and miscarriage. Identify the disease then explain how the parasite can enter the human body and how one can prevent spread of this parasite. (give six points) **(15 marks)**

11. You soak juicy grapes in a bowl of concentrated sugar, water after a while you notice the grapes shrink and wrinkle, while the raisins you added previously pumps up and again their size. Explain what happening at the cellular level with osmosis, diffusion and mass flow. **(15 marks)**